

Internship Guide – Predictive Modeling

Contents

1	Context and Positioning	3
1.1	Gigalis and the Digital Central Purchasing Body	3
1.2	The Data & AI Hub and the Data Control Tower	3
1.3	The Problem Addressed by the Internship	3
2	Scientific Foundations and State of the Art	4
2.1	Survival Analysis	4
2.1.1	Fundamental Concepts	4
2.1.2	Bibliographic References – Survival Analysis	5
2.2	Change-Point Detection and Trend Analysis	6
2.2.1	Fundamental Concepts	6
2.2.2	Bibliographic References – Time Series	6
2.3	Natural Language Processing for Calls for Tenders	7
2.3.1	Fundamental Concepts	7
2.3.2	Bibliographic References – NLP	8
2.4	Causal Inference: Internship Perspective	9
2.4.1	Why Prediction Is Not Enough	9
2.4.2	Bibliographic References – Causal Inference	9
3	Data Sources and Acquisition Strategy	10
3.1	Overview of Sources	10
3.2	Corpus Construction Strategy	11
3.2.1	The Problem of Linking Renewals	11
3.2.2	Recommended Scope for the Internship	11
3.3	Data Quality and Treatment of Missing Values	12
4	Detailed Work Plan – 12 Weeks	12

4.1	Phase 1 – Data Acquisition and Qualification, S1–S3	13
4.1.1	Week 1: Exploration and Mapping of Sources	13
4.1.2	Week 2: Cleaning and Filtering	13
4.1.3	Week 3: Renewal Linking and Data Quality Report	14
4.2	Phase 2 – NLP Classification of Digital Contracts, S4–S5	14
4.2.1	Week 4: Annotation and Baseline	14
4.2.2	Week 5: Final Model and Propagation	14
4.3	Phase 3 – Survival Modeling, S6–S8	15
4.3.1	Week 6: Exploratory Survival Analysis	15
4.3.2	Week 7: Cox Model and Assumption Validation	15
4.3.3	Week 8: Parametric Models and Individual Predictions	15
4.4	Phase 4 – Trend and Change-Point Detection, S9–S10	16
4.4.1	Week 9: Construction of Time Series	16
4.4.2	Week 10: Change-Point Detection and Interpretation	16
4.5	Phase 5 – Synthesis and Delivery, S11–S12	16
4.5.1	Week 11: Writing and Integration	16
4.5.2	Week 12: Defense and Transfer	17
5	Expected Deliverables and Evaluation Criteria	17
5.1	List of Deliverables	17
5.2	What Is Not Expected	17
	Translation Choices Requiring Human Review	18

1 Context and Positioning

1.1 Gigalis and the Digital Central Purchasing Body

Gigalis is the public interest group for digital services in the Pays de la Loire region. As a public operator, Gigalis carries out several complementary missions: connectivity, sovereign cloud services, support for the cybersecurity of regional public-sector actors, and the role of a digital central purchasing body (*centrale d'achat numérique*) for public institutions in France.

As a central purchasing body, Gigalis designs and manages pooled framework agreements (*accords-cadres*) in the areas of cloud services, cybersecurity, digital infrastructure, and artificial intelligence. These contracts are made available to members without any obligation to use them. The strategic relevance of these contracts depends directly on Gigalis's ability to anticipate the future needs of public-sector actors. This internship aims to provide tools for that purpose.

1.2 The Data & AI Hub and the Data Control Tower

Gigalis is developing a Data & AI Hub whose ambition is to transform internal and open data into decision-support capabilities. The Data Control Tower is the target vision: an integrated steering system that makes it possible to anticipate changes in the regional and national public digital ecosystem.

This internship is the first operational building block of that vision: building the models that will make it possible to move from a descriptive logic, “what happened?”, to a predictive logic, “what will happen?”.

1.3 The Problem Addressed by the Internship

The central strategic question that this internship should help address is the following:

Central question

How can we estimate, using historical data on public digital procurement, the probability that a contract or a technological segment will generate an identifiable purchasing need within the next 12 to 24 months, and how can this estimate inform the strategy of Gigalis's central purchasing body?

This question can be broken down into three distinct sub-problems, each associated with a family of mathematical methods:

- **The lifetime problem:** how much time typically elapses between the launch of a public digital contract and its renewal? This problem falls under survival analysis.
- **The trend problem:** which technological segments are growing, plateauing, or declining? This problem falls under change-point detection in time series.
- **The text-signal problem:** how can the technological theme of a call for tenders written in natural language be extracted automatically? This problem falls under natural language processing.

The response to these three sub-problems is the core of the internship. If time allows, a fourth

dimension, causal inference, will be addressed as a research perspective and outline for the internship.

2 Scientific Foundations and State of the Art

This section presents the theoretical foundations of the approaches used in this internship. For each area, an introduction to the fundamental concepts is provided, followed by the main bibliographic references that the intern will need to master before starting the modeling work.

2.1 Survival Analysis

2.1.1 Fundamental Concepts

Survival analysis is the statistical framework dedicated to modeling the time elapsed until an event of interest occurs. In the context of this internship, the event is the renewal, or abandonment, of a public digital contract. The time considered is the duration between the notification of the contract and its official re-launch.

The key feature of this framework is censoring: some contracts are still active at the observation date, and their true lifetime is unknown. Ignoring this censoring, by treating these contracts as if they had not yet been renewed, introduces systematic bias into the estimation. Survival analysis handles censoring in a rigorous way.

The survival function $S(t)$ and the Kaplan-Meier estimator The survival function

$$S(t) = P(T > t)$$

represents the probability that a contract survives, meaning that it remains active without renewal, beyond time t . The Kaplan-Meier estimator is the standard non-parametric estimator for $S(t)$. It is non-parametric, which means that it makes no assumption about the shape of the survival distribution. This is its main strength for exploratory analysis.

The intern should be able to plot and interpret Kaplan-Meier curves stratified by technological segment, such as cloud, cybersecurity, infrastructure, and AI, and by buyer type, such as local authority, public health institution, and educational institution.

The Cox proportional hazards model The Cox model makes it possible to estimate the effect of covariates on the risk of renewal without specifying the shape of the baseline hazard function. This is the semi-parametric assumption. The model is written as:

$$h(t | X) = h_0(t) \cdot \exp(\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p),$$

where $h_0(t)$ is the baseline hazard function, which is not estimated parametrically, and the β_i are the coefficients associated with the covariates X_i , such as contract amount, technological segment, initial duration, buyer type, and region.

The proportional hazards assumption, which states that the hazard ratio between two individuals is constant over time, must be tested using the Schoenfeld residuals test. If the assumption is

violated, either stratification or interactions with time must be introduced.

Parametric models: Weibull, log-logistic, Gompertz When the objective is to make individual predictions, for example the probability that a specific contract will be renewed within 12 months, parametric models are more suitable than the Cox model because they allow extrapolation beyond the observation period. The Weibull model is especially relevant: its shape parameter κ makes it possible to model increasing hazards, $\kappa > 1$, where aging contracts are more likely to be renewed; constant hazards, $\kappa = 1$, corresponding to the exponential distribution; or decreasing hazards, $\kappa < 1$.

Shared frailty models Contracts from the same buyer are not independent. A buyer that renews its contracts quickly tends to do so systematically. This intra-group dependence violates the classical independence assumption between observations. Shared frailty models introduce an unobserved random effect, the frailty, at the buyer level in order to capture this heterogeneity. This is the level of sophistication that would distinguish a master’s thesis from a standard analysis.

2.1.2 Bibliographic References – Survival Analysis

ID	Reference and relevance
[R1]	Cox, D.R. (1972). <i>Regression Models and Life-Tables</i> . Journal of the Royal Statistical Society, Series B, 34(2), 187–220. Foundational article for the Cox model. Required reading, at least the introduction and Section 2. Cox’s partial likelihood method is one of the most elegant contributions to twentieth-century statistics.
[R2]	Kaplan, E.L. & Meier, P. (1958). <i>Nonparametric Estimation from Incomplete Observations</i> . Journal of the American Statistical Association, 53(282), 457–481. Original article introducing the Kaplan-Meier estimator. An essential historical reading. More than 60,000 citations, making it one of the most cited articles in statistics.
[R3]	Kalbfleisch, J.D. & Prentice, R.L. (2002). <i>The Statistical Analysis of Failure Time Data</i> (2nd ed.). Wiley Series in Probability and Statistics. The key reference for parametric and semi-parametric survival analysis. Chapters 1–4 cover the needs of this internship.
[R4]	Therneau, T.M. & Grambsch, P.M. (2000). <i>Modeling Survival Data: Extending the Cox Model</i> . Springer, New York. The most practical book on the Cox model, including tests of the proportional hazards assumption and extensions such as frailty and time dependence. The R package <code>survival</code> is documented here.
[R5]	Fazekas, M. & Kocsis, G. (2020). <i>Uncovering High-Level Corruption: Cross-National Objective Corruption Risk Indicators Using Public Procurement Data</i> . Political Research Quarterly, 73(1), 155–177. Concrete application of large-scale public procurement data analysis. Provides insights into the qualitative limitations of BOAMP/TED data and cleaning strategies.

ID	Reference and relevance
[R6]	Wager, S. & Athey, S. (2018). <i>Estimation and Inference of Heterogeneous Treatment Effects using Random Forests</i> . Journal of the American Statistical Association, 113(523), 1228–1242. For further study: causal forests for estimating the heterogeneous effect of covariates on survival.

2.2 Change-Point Detection and Trend Analysis

2.2.1 Fundamental Concepts

Public procurement time series have characteristics that make classical methods, such as ARIMA and exponential smoothing, difficult to apply directly: they are short, typically 5 to 10 years of monthly or quarterly data; irregular; potentially non-stationary; and subject to structural breaks caused by policy changes, crises, or technological transformations.

The objective of this part of the internship is not to forecast the future values of the series, which would require much longer series, but to detect and date breaks in trends. In other words, the goal is to identify the moments when the behavior of a technological segment changes structurally.

Regime-switching models: Hidden Markov Models A Hidden Markov Model, or HMM, assumes that the time series is generated by a hidden process that alternates between several states, or regimes: strong growth, plateau, and decline. At each point in time, the probability of moving from one regime to another is described by a transition matrix. The Viterbi algorithm makes it possible to estimate the most likely sequence of states. This approach is especially suited to our context because it does not assume that transitions are rare.

Change-point detection: PELT, CUSUM, BOCPD The PELT algorithm, Pruned Exact Linear Time, introduced by Killick et al. (2012), determines the number and positions of breaks in a series by minimizing a penalized criterion, such as BIC or mBIC. It is the reference algorithm for offline change-point detection. The Python package `ruptures`, described by Truong et al. (2020), provides a robust implementation.

Online detection, BOCPD or Bayesian Online Changepoint Detection, introduced by Adams & MacKay (2007), is more appropriate if new data are to be integrated progressively. It maintains a probability distribution over the time elapsed since the last break, which allows near-real-time alerts.

Local trend models: Local Level Model, TBATS For series with seasonality, such as tendering cycles at the end of the budget year, and non-stationary trends, state space models and their special case, the Local Level Model, allow a rigorous trend-season-residual decomposition. TBATS, introduced by De Livera et al. (2011), handles multiple and complex seasonalities.

2.2.2 Bibliographic References – Time Series

ID	Reference and relevance
[R7]	Killick, R., Fearnhead, P. & Eckley, I.A. (2012). <i>Optimal Detection of Changepoints With a Linear Computational Cost</i> . Journal of the American Statistical Association, 107(500), 1590–1598. Foundational article for the PELT algorithm. Required reading before using the <code>ruptures</code> package. It is important to understand the penalty and its practical implications.
[R8]	Adams, R.P. & MacKay, D.J.C. (2007). <i>Bayesian Online Changepoint Detection</i> . arXiv:0710.3742 [stat.ML]. Foundational article for BOCPD. Short, accessible, and mathematically elegant. The online Bayesian approach is the most suitable for a continuous monitoring context.
[R9]	Truong, C., Oudre, L. & Vayatis, N. (2020). <i>Selective review of offline change point detection methods</i> . Signal Processing, 167, 107299. Systematic and comparative review of offline change-point detection methods. An ideal starting point for choosing the method suited to BOAMP data.
[R10]	De Livera, A.M., Hyndman, R.J. & Snyder, R.D. (2011). <i>Forecasting Time Series With Complex Seasonal Patterns Using Exponential Smoothing</i> . Journal of the American Statistical Association, 106(496), 1513–1527. Article introducing TBATS. Relevant if procurement series show seasonalities linked to annual budget cycles.
[R11]	Hamilton, J.D. (1989). <i>A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle</i> . Econometrica, 57(2), 357–384. Foundational article for Markov-switching models. Demanding but essential for understanding HMMs applied to economic time series.

2.3 Natural Language Processing for Calls for Tenders

2.3.1 Fundamental Concepts

The BOAMP/TED text corpus is composed of highly standardized legal-administrative texts. This standardization is both a constraint, because there is limited lexical variation, and an opportunity, because technological themes are often explicitly named. NLP in this context mainly aims to automate the classification of contracts by technological theme, which is a prerequisite for all quantitative analyses.

Bag of Words model and TF-IDF The TF-IDF representation, Term Frequency-Inverse Document Frequency, is the natural starting point for this corpus. It assigns each term in each document a weight reflecting its relative importance within that document compared with the full corpus. For tender texts, a manually curated dictionary of technological terms, or whitelist, combined with TF-IDF may be sufficient for reliable initial classification.

Topic modeling: LDA and its extensions Latent Dirichlet Allocation, introduced by Blei et al. (2003), is a generative probabilistic model that assumes each document is a mixture of

latent topics, and each topic is a distribution over the vocabulary. Inference is performed using variational EM or MCMC. For this internship, LDA could make it possible to discover thematic groupings without prior supervision.

The Dynamic LDA extension, introduced by Blei & Lafferty (2006), models the temporal evolution of topics: distributions over the vocabulary evolve from period to period according to a Wiener process. This extension is directly relevant for tracking the emergence of new technologies. For example, the “AI” topic should emerge in the years 2015–2018 and grow strongly after 2020.

Transformers and contextual embeddings Models from the BERT family, introduced by Devlin et al. (2019), and their French-language derivatives, such as CamemBERT and DéBERTa-fr, make it possible to obtain contextual vector representations of texts that capture semantics beyond simple word counts. For classification, a CamemBERT model fine-tuned on a small set of annotated examples, a few hundred manually labeled calls for tenders, will produce markedly better results than TF-IDF+SVM for ambiguous technological categories.

Caution: fine-tuning BERT requires a GPU. If the working environment does not have a GPU, BERT embeddings can be used as fixed features, for example with `sentence-transformers`, without fine-tuning. This works well in zero-shot inference for clearly defined categories.

2.3.2 Bibliographic References – NLP

ID	Reference and relevance
[R12]	Blei, D.M., Ng, A.Y. & Jordan, M.I. (2003). <i>Latent Dirichlet Allocation</i> . Journal of Machine Learning Research, 3, 993–1022. Foundational article for LDA. Required reading. The article is remarkably clear in presenting a generative probabilistic model. Section 1, the introduction, and Section 2, the model, are sufficient for this internship.
[R13]	Blei, D.M. & Lafferty, J.D. (2006). <i>Dynamic Topic Models</i> . Proceedings of the 23rd International Conference on Machine Learning (ICML), 113–120. Temporal extension of LDA. Directly applicable to tracking the evolution of technological topics in the BOAMP corpus over 10 years.
[R14]	Devlin, J., Chang, M-W., Lee, K. & Toutanova, K. (2019). <i>BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding</i> . Proceedings of NAACL-HLT 2019, 4171–4186. Foundational article for BERT. The Transformer architecture, introduced by Vaswani et al. (2017), should be understood in parallel. For this internship, CamemBERT, introduced by Martin et al. (2020), should be used rather than English BERT.
[R15]	Reimers, N. & Gurevych, I. (2019). <i>Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks</i> . Proceedings of EMNLP-IJCNLP 2019, 3982–3992. Practical approach for obtaining efficient sentence embeddings without fine-tuning. The <code>sentence-transformers</code> library can be used directly for this internship.

ID	Reference and relevance
[R16]	Martin, L. et al. (2020). <i>CamemBERT: a Tasty French Language Model</i> . Proceedings of ACL 2020, 7203–7219. BERT model for French. Required reference for any NLP task on French tender texts. Available on HuggingFace as <code>camembert-base</code> .

2.4 Causal Inference: Internship Perspective

2.4.1 Why Prediction Is Not Enough

A predictive model can answer the question “what will happen?”. Causal inference can answer the question “what would happen if Gigalis did X?”. For the central purchasing body, the central causal question is the following: does the implementation of a pooled framework agreement by Gigalis actually change the purchasing behavior of local authorities, in terms of amounts, frequencies, and segments, or does Gigalis simply capture demand that would have existed anyway?

This question cannot be answered by a purely descriptive or predictive analysis. It requires an explicit causal framework.

The potential outcomes framework: Rubin Causal Model In the Rubin framework, introduced in 1974, each local authority that is a member of Gigalis has two potential outcomes: the purchasing behavior it would have with the pooled contract, and the purchasing behavior it would have without it. The fundamental problem of causal inference is that both cannot be observed simultaneously. This is the counterfactual problem. The question then becomes: how can a credible control group be constructed, made up of comparable local authorities that did not use the pooled contract?

Causal identification methods Depending on the structure of the available data, several methods can be considered:

- **Difference-in-differences (DiD):** compare the evolution of members before and after they joined Gigalis with the evolution of comparable non-member local authorities. The key assumption is the prior parallel trends assumption.
- **Propensity score matching (PSM):** construct a synthetic control group of non-members that are statistically comparable to members on observable covariates, such as size, budget, and type of local authority.
- **Regression discontinuity design (RDD):** if a threshold rule determines eligibility for a Gigalis contract, for example a population threshold, local authorities just above and just below the threshold form a natural quasi-experiment.

2.4.2 Bibliographic References – Causal Inference

ID	Reference and relevance
[R17]	Angrist, J.D. & Pischke, J-S. (2009). <i>Mostly Harmless Econometrics: An Empiricist's Companion</i>. Princeton University Press. The most readable textbook on causal inference in econometrics. Chapters 3, on instrumental variables, and 5, on DiD, are directly relevant. To be read alongside the work on BOAMP data in order to conceptualize an identification strategy.
[R18]	Pearl, J., Glymour, M. & Jewell, N.P. (2016). <i>Causal Inference in Statistics: A Primer</i>. Wiley. Introduction to the DAG framework, Directed Acyclic Graphs, for representing causal assumptions. More formal than Angrist & Pischke, but essential for understanding which assumptions are being made implicitly.
[R19]	Athey, S. & Imbens, G.W. (2017). <i>The State of Applied Econometrics: Causality and Policy Evaluation</i>. Journal of Economic Perspectives, 31(2), 3–32. Accessible review article on modern causal methods, including causal forests and causal machine learning.
[R20]	Callaway, B. & Sant'Anna, P.H.C. (2021). <i>Difference-in-differences with multiple time periods</i>. Journal of Econometrics, 225(2), 200–230. Extension of classical DiD to contexts with staggered adoption over time, which is exactly the situation of Gigalis members joining the central purchasing body at different dates.
[R21]	Fazekas, M. & Blumenstock, G. (2020). <i>One for All? Pathways from Single to Multi-Level Disclosure in Beneficial Ownership Transparency Policies</i>. OpenOwnership, Research Report. Empirical study on the impact of transparency policies on public procurement. DiD methodology applied to European public procurement data, a case very close to ours.

3 Data Sources and Acquisition Strategy

3.1 Overview of Sources

This internship relies on open public procurement data. Before starting the modeling work, the intern will need to devote significant time to understanding the structure, quality, and limitations of these sources. Underestimating this work is the main risk of failure for the internship.

Source	Description	Format / Access	Main limitations
BOAMP	<i>Bulletin Officiel des Annonces des Marchés Publics</i> . Contract notices, public call for competition notices (AAPC), and award notices for French public contracts.	REST JSON API + Open Data, annual files.	Variable CPV quality. Amounts often missing or incorrect. No automatic linking of renewals.

Source	Description	Format / Access	Main limitations
TED	Tenders Electronic Daily. Official journal of European public procurement, covering contracts above EU thresholds (JOUE).	TED API + bulk CSV/XML data.	Only contracts above European thresholds. Over-representation of large buyers.
data.gouv.fr	Open datasets from the French state. Includes DECP data, <i>Données Essentielles de la Commande Publique</i> .	Downloadable CSV/JSON. DECP format standardized since 2018.	Incomplete coverage before 2019. Limited granularity on contract objects.
FOPPA	French Open Public Procurement Award Notices. Structured dataset built from BOAMP award notices.	Enriched CSV, Open Data access.	Mainly covers award notices, not contract notices. Limited time scope.

3.2 Corpus Construction Strategy

3.2.1 The Problem of Linking Renewals

The main technical difficulty of this internship is the following: BOAMP data do not provide an explicit link between a contract and its later renewal. This linking must be constructed manually using matching rules based on:

- the buyer identifier, SIRET or SIREN;
- textual similarity of the contract object, using cosine distance on embeddings or TF-IDF;
- temporal proximity, since a four-year contract is typically renewed around plus or minus six months from its expected end date;
- the CPV code, corresponding to the product or service category.

This linking will never be perfect. The intern will need to quantify the rate of missed matches and assess the impact of the resulting selection bias on survival models. A linking rate of 40–60% is an acceptable result on this database.

3.2.2 Recommended Scope for the Internship

Working on all French public procurement would be counterproductive for a three-month internship. The recommended scope is the following:

Recommended working scope

Geography: Pays de la Loire as the priority area, with the Grand Ouest as an extension if the volume is insufficient for survival models.

Themes: digital contracts identified by CPV codes 48, software; 72, IT services; 32, telecommunications equipment; 35, security; and their subcategories.

Period: 2015–2024, meaning 10 years of data, allowing at least one full renewal cycle to be observed for four-year contracts.

Target volume: 2,000 to 5,000 contracts after filtering and cleaning. Below 1,000 contracts, survival models will lack statistical power.

3.3 Data Quality and Treatment of Missing Values

BOAMP data suffer from several quality issues documented in the literature, including Fazekas & Kocsis (2020):

- poorly filled or generic CPV codes, for example CPV 72000000, “IT services”, without a subcategory;
- aberrant amounts, such as 0 EUR or ceiling amounts that are not representative of the true value;
- missing contract durations or durations expressed in whole years without precision;
- inconsistent buyer names, such as missing SIREN identifiers or multiple legal names for the same organization.

The intern must produce a data quality report by the end of Week 3, before starting any modeling work. This report must precisely quantify missing-value rates by variable, the distributions of key variables, and the cleaning rules applied.

4 Detailed Work Plan – 12 Weeks

The work plan is organized into four phases. Each phase produces an intermediate deliverable. Phase 4, causal inference, is optional and conditional on the satisfactory completion of Phases 1 to 3.

Weeks	Phase	Main work	Deliverables
S1–S3	Phase 1 – Data and corpus	BOAMP/TED API acquisition, cleaning, renewal linking, enriched CPV taxonomy, data quality report.	Cleaned corpus + data quality report.
S4–S5	Phase 2 – NLP classification	Manual annotation of a sample, 300–500 contracts, training of a CamemBERT/TF-IDF classifier, evaluation, propagation to the whole corpus.	Annotated corpus + documented classification model.
S6–S8	Phase 3 – Survival models	Exploratory duration analysis, Kaplan-Meier, Cox model with assumption testing, Weibull model for individual predictions.	Survival analysis notebook + results report.

Weeks	Phase	Main work	Deliverables
S9–S10	Phase 4 – Trends and change points	Construction of time series by segment, application of PELT or BOCPD, interpretation of breaks, visualization.	Trend detection report with visualizations.
S11–S12	Phase 5 – Synthesis and delivery	Final methodological report, demonstration notebook, internal defense, pipeline documentation.	Final report + notebook + presentation.

4.1 Phase 1 – Data Acquisition and Qualification, S1–S3

4.1.1 Week 1: Exploration and Mapping of Sources

Objective: understand precisely what BOAMP and TED data contain before writing any modeling code.

- Access the BOAMP API, official documentation: `api.boamp.fr`. Download a sample of 500 notices to understand the JSON structure.
- Identify the available fields: buyer identifier, contract object, CPV codes, amount, duration, notification date, award date.
- Consult the DECP dataset available on `data.gouv.fr` and compare it with BOAMP: which variables are better completed in which source?
- Produce a one-page report covering the data structure, available fields, estimated completion rate, and decision on the main source.

Scientific question of the week: to what extent is the “contract duration” variable reliable in BOAMP data? How does the declared duration compare with the observed duration, defined as the delay between notification and identifiable renewal?

4.1.2 Week 2: Cleaning and Filtering

Objective: build a reproducible and documented cleaning pipeline.

- Define filtering rules to identify digital contracts, using a list of relevant CPV codes and a list of keywords in the contract object.
- Clean amounts, including aberrant values, 0 EUR values, and ceiling amounts. Define a strategy for missing values: median imputation by segment or exclusion?
- Normalize buyer identifiers, SIREN/SIRET. Use the SIRENE API to enrich buyer data, including type of local authority and region.
- Build a taxonomic reference system: 8 to 12 technological categories with their CPV mappings and characteristic keywords.

4.1.3 Week 3: Renewal Linking and Data Quality Report

Objective: build the target variable for survival analysis: time until renewal.

- Implement the linking algorithm: for each notified contract, search for a later contract from the same buyer, in the same segment, within a defined time window.
- Evaluate the linking rate. Analyze potential biases: are contracts with no identifiable successor systematically different from the others?
- Produce the data quality report: a 3- to 5-page document presenting descriptive statistics for the final corpus, cleaning rules and their impacts, and identified limitations.

Phase 1 deliverable – Acceptance criteria

The cleaned corpus must contain at least 800 contracts with a documented observed or censored duration.

The data quality report must precisely quantify linking rates by segment and identify potential biases.

The cleaning pipeline must be fully reproducible, with documented Python scripts and no untracked manual manipulation.

4.2 Phase 2 – NLP Classification of Digital Contracts, S4–S5

4.2.1 Week 4: Annotation and Baseline

Objective: build a small annotated dataset and establish a classification baseline.

- Manually annotate 300 to 500 contracts by assigning them to the taxonomy defined in Phase 1. Annotation must be carried out by at least two annotators, the intern and the supervisor, in order to calculate inter-annotator agreement, Cohen’s kappa.
- Train a first baseline model: TF-IDF + logistic regression or SVM. Evaluate it using stratified cross-validation, with macro F1-score.
- Analyze errors: which types of contracts are most often misclassified? Why? Review the taxonomy if necessary.

4.2.2 Week 5: Final Model and Propagation

Objective: improve the classifier and propagate labels to the whole corpus.

- Test CamemBERT, either fine-tuned on 300 annotated examples, or **sentence-transformers** with pre-computed embeddings plus a linear classifier. Compare the result with the baseline.
- Analyze the marginal contribution of the advanced model compared with TF-IDF+SVM. If the gain is < 5 F1 points, the baseline is sufficient.
- Propagate labels to the whole corpus. For each document, provide not only the predicted class but also the confidence probability, which is essential in order not to introduce noise into later analyses.

- Filter low-confidence predictions, < 0.7 , for trend analyses. Keep only well-classified contracts.

4.3 Phase 3 – Survival Modeling, S6–S8

4.3.1 Week 6: Exploratory Survival Analysis

Objective: understand the empirical structure of durations before fitting any model.

- Build global Kaplan-Meier curves and curves stratified by technological segment, buyer type, such as local authority, public institution, or other, and amount bracket.
- Test differences between curves using the log-rank test. Formulate the statistical hypotheses in operational terms: are cloud contracts renewed significantly faster than infrastructure contracts?
- Analyze censoring: what proportion of contracts is censored? Is censoring informative, meaning linked to contract characteristics, or independent?
- Identify the segments for which the data are too limited for reliable analysis. Empirical rule: at least 30 events per stratum for Kaplan-Meier, and 10 per covariate for Cox.

4.3.2 Week 7: Cox Model and Assumption Validation

Objective: estimate the effect of covariates on the renewal hazard while respecting the necessary statistical rigor.

- Fit a univariate Cox model for each candidate covariate. Select covariates for the multivariate model using AIC and the Wald test.
- Test the proportional hazards assumption using Schoenfeld residuals and the Grambsch-Therneau test. In case of violation, stratify by the problematic variable or introduce an interaction with time.
- Interpret hazard ratios with their 95% confidence intervals. Beware of overfitting: limit the model to one covariate per 10 observed events, following the rule of Van Belle et al. (2002).
- Validate the model using temporal cross-validation, training on 2015–2021 and evaluating on 2022–2024. Calculate Harrell’s concordance index C .

4.3.3 Week 8: Parametric Models and Individual Predictions

Objective: produce predicted probabilities of renewal at 12 and 24 months for active Gigalis contracts.

- Fit parametric models, including Weibull, log-logistic, and generalized gamma. Compare AIC values. Check goodness of fit graphically using linearized survival plots.
- For each active contract in the Gigalis portfolio, produce an estimate of $\Pr(\text{renewal within 12 months})$ and $\Pr(\text{renewal within 24 months})$, with confidence intervals.

- Build a dashboard in notebook format, Jupyter, presenting the 20 contracts with the highest probability of renewal at 12 months. This is the first operational deliverable of the internship.

4.4 Phase 4 – Trend and Change-Point Detection, S9–S10

4.4.1 Week 9: Construction of Time Series

Objective: build and analyze time series of activity by technological segment.

- For each segment in the taxonomy, build a monthly or quarterly time series for the number of notified contracts, total amount, and median duration.
- Visualize the series. Visually identify anomalies, jumps, and trends. Annotate known exogenous events, such as COVID, the Military Programming Law, and the national AI strategy.
- Test stationarity using the ADF and KPSS tests. Identify the required transformations, such as differencing or log transformation for amounts.

4.4.2 Week 10: Change-Point Detection and Interpretation

Objective: apply change-point detection algorithms and interpret the results in operational terms.

- Apply the PELT algorithm, using the `ruptures` package, with BIC penalty on amount series by segment. Test sensitivity to the penalty parameter.
- For high-volume segments, apply a three-state HMM, growth/plateau/decline, and estimate the probability of being in each regime at the current date.
- Produce an interpretation report: for each detected break, formulate a hypothesis about its cause, such as regulatory change, technological emergence, or exogenous event. Validate this with the supervisor.
- Build a signal matrix: for each segment, one row summarizing the estimated current trend, the date of the last break, and the recommendation for the central purchasing body.

4.5 Phase 5 – Synthesis and Delivery, S11–S12

4.5.1 Week 11: Writing and Integration

- Write the final methodological report. Recommended structure: context, data, methods, with one section per family of methods, results, limitations, and perspectives.
- Document the entire data and modeling pipeline. Each script must be commented, versioned with Git, and accompanied by a README.
- Prepare the demonstration notebook: it must be executable end to end by someone who did not participate in the internship.

4.5.2 Week 12: Defense and Transfer

- Internal 30-minute presentation to the Gigalis team, followed by 20 minutes of questions.
- Transfer of deliverables to the Gigalis environment, including internal Git repository, documentation, and consolidated data.
- Draft a one-page executive summary for management: what was done, what works, what remains uncertain, and the recommended next steps.

5 Expected Deliverables and Evaluation Criteria

5.1 List of Deliverables

#	Deliverable	Description	Delivery week
L1	Data quality report	Descriptive statistics of the corpus, linking rate, identified biases, documented cleaning rules.	S3
L2	Annotated corpus + NLP model	300–500 annotated contracts, evaluated classification model, F1 score and confusion matrix, propagation to the whole corpus.	S5
L3	Survival analysis notebook	KM curves, validated Cox model, Weibull model, individual predictions at 12/24 months, table of contracts with high renewal risk.	S8
L4	Trend report	Time series by segment, detected and interpreted break points, signal matrix for the central purchasing body.	S10
L5	Final methodological report	40- to 60-page synthesis document covering the full approach, methodological choices, results, and limitations.	S12
L6	Documented pipeline, Git repository	Python code, README, requirements.txt , execution instructions. Complete reproducibility.	S12

5.2 What Is Not Expected

Deliverables excluded from the scope of this internship

- Interactive web dashboard or front-end application.
- Precise absolute forecasts: the objective is probabilistic indicators and trends, not point predictions.
- Exhaustive national coverage: the regional scope is assumed and justified.
- Complex deep learning models: methodological rigor on well-understood models takes priority over the exploration of complex architectures.

Translation Choices Requiring Human Review

- *Centrale d'achat* was translated as “central purchasing body” and the French term was kept in parentheses at first use.
- *Marché* was generally translated as “contract” or “public contract”, rather than “market”, because the document concerns public procurement.
- *Collectivité* was translated as “local authority”. Depending on the intended audience, “local government body” could also be acceptable.
- *Avis d'attribution* was translated as “award notice”, and AAPC was kept with an explanatory translation.
- The source text appears to contain a possible inconsistency: the work plan says it is organized into four phases and mentions “Phase 4, causal inference”, but the table lists five phases and uses Phase 4 for trends and change points. This translation preserves the source structure and wording.